MEDI-CAPS UNIVERSITY, INDORE

Faculty of Engineering • Department of Computer Science & Engineering

Academic Session 2026 – 2027

CAPSTONE PROJECT 1 • TECHNICAL REPORT

MULTIMODAL CONSTRUCTION INTELLIGENCE PLATFORM

Autonomous Vision Auditing, Forensic Crack Diagnostics, Real Estate Valuation & Weather-Aware Decision Support

Domain: Generative AI & Decision Support	**Platform ID:** 6ac4c98d1dcde6d7857c1642
Core Vision Engine: Google Gemini 1.5 Multimodal	**Generative 3D:** Pollinations AI / Flux API
Governing Standards: IS 456:2000, IS 10262, IS 13920	**Safety Regulations:** OSHA 1926, IS 2925, IS 15809

SUBMITTED TO:	SUBMITTED BY:
Prof. Shruti Mam Professor, Department of Computer Science & Engineering Course: Generative Artificial Intelligence **Medi-Caps University**, Indore (M.P.), India	**Sulabh Kumar Sharma** Enrollment No: **EN23CS304074** B.Tech Computer Science & Engineering (Gen AI) **Medi-Caps University**, Indore (M.P.), India

■ **GitHub Repository:** https://github.com/sulabh1-y/Multimodal-Construction-Intelligence-Platform

■ **Cloud Ready:** Configured for 1-Click Streamlit Community Cloud Deployment (Zero External DB Dependencies)

1. Executive Summary & Problem Formulation

Modern infrastructure projects face high failure rates, unsafe working conditions, structural cracking defects, cost overruns, and severe weather disruptions. According to the International Labour Organization (ILO) and India's Directorate General of Factory Advice Service and Labour Institutes (DGFASLI), over 24% of occupational fatalities in construction are attributed to fall-from-height incidents and non-compliance with Personal Protective Equipment (PPE). Simultaneously, unmonitored concrete cracking causes progressive structural shear failures requiring millions in remediation.

The **Multimodal Construction Intelligence Platform (MCIP)** solves these challenges by combining multimodal computer vision (**Google Gemini 1.5 Flash / Pro**), strict JSON data contracts, and Indian Standard civil codes (**IS 456:2000, IS 10262, IS 13920, IS 2925, IS 15809** and **OSHA 1926**) into a streamlined real-time decision-support system.

Table 1: Kanban Epics Architecture & Scope

Epic ID	Module Name	Governing Standard	Key Deliverables & Responsibilities
Epic 1	Requirements & Environment	Python 3.11 / Streamlit	Full environment setup, zero-database dependencies, graceful offline fal
Epic 2	PPE Vision & Crack Diagnostics	IS 2925 / IS 15809 / OSHA 1926 IS 456:2000 / ACI 224R	Automated safety score (0-100), hazard flagging, crack aperture (mm), re
Epic 3	Planning & Real Estate Valuation	CPWD Norms / Indian City Tiers	Total budget, CapEx incurred, WIP valuation (INR Cr/L), material forecas
Epic 4	Multimodal AI & JSON Contracts	Pydantic / RFC 8259	Schema-validated JSON contracts for site safety audits and structural cra
Epic 5	Weather Advisory & Site Chatbot	IS 7861 / OSHA 1926.1431 IS 456 Table 11 & Table 16	Meteorological hazard mitigation matrix, multi-turn AI civil engineering as
Epic 6	Testing & Cloud Deployment	Streamlit Cloud / GitHub CI	End-to-end 7-stage automated verification suite passing 100% test cover

Core System Objectives:

• **Automate Site Safety Audits:** Detect compliance with helmets, high-vis vests, steel-toe boots, and fall-arrest harnesses under OSHA 1926 and IS 2925.

• **Forensic Crack Classification:** Classify concrete fractures into structural shear, hairline, settlement, thermal expansion, or spalling under IS 456:2000 Clause 40.

• **Econometric Asset Valuation:** Calculate CapEx incurred, ready market values in INR (Crores & Lakhs), and project materials (Cement, TMT Steel, Sand, Aggregate).

• **Dynamic Weather Mitigation:** Enforce OSHA tower crane halts (>35 km/h) and IS 7861 hot weather concreting protocols.

2. System Architecture & Multimodal Pipeline

MCIP operates on a layered, decoupled architecture ensuring sub-second UI responsiveness, strict data validation, and resilient fallback modes when operating in bandwidth-constrained jobsite environments.

Layer	Technology Stack	Operational Responsibilities
Presentation Layer	Streamlit 1.57, Plotly Express, Custom CSS	Interactive 4-tab interface, metric cards, status badges, dynamic charts, file uploaders, and 1-click JSON export.
Multimodal Vision Layer	Google Gemini 1.5 Flash / Pro, Pillow	Ingests site photography, processes zero-shot civil engineering prompts, enforces JSON schema output at temperature 0.2.
Offline Simulation Engine	Python Civil Benchmarks, IS 456 Matrix	Enables complete, high-fidelity demonstration simulation if no API key is supplied; conforms strictly to verified benchmarks.
Valuation & Material Engine	CPWD Cost Norms, IS 456 Thumb Rules	Computes CapEx incurred, WIP valuation, and exact quantities of Cement bags, TMT Steel MT, Sand and Aggregate in Brass.
Generative Rendering Engine	Pollinations AI / Flux API	Synthesizes architectural style presets, lighting conditions, and camera angles into photorealistic 3D conceptual renders.
Safety & Weather Matrix	OSHA 1926.1431, IS 7861, IS 456 Table 11	Enforces dynamic meteorological directives and powers the multi-turn SiteVision AI civil engineering assistant chatbot.

End-to-End Multimodal Vision Data Flow:

1. **Image Ingestion:** User uploads a site inspection photo or selects a pre-loaded benchmark.
2. **Preprocessing:** Pillow converts image to RGB, resizes and encodes into Base64 JPEG.
3. **Zero-Shot Prompting:** Gemini 1.5 receives strict system instructions demanding RFC 8259 JSON.
4. **Contract Validation:** System parses candidate text, validates fields (`safety_score`, `ppe_detection`, `hazards`), and renders metric cards.
5. **Reporting:** User inspects findings and downloads the standardized JSON compliance contract.

3. Tab 1: Site Safety & PPE Compliance Audit (Epic 2)

Worker safety compliance is evaluated against OSHA 1926 regulations and Indian Standards (IS 2925, IS 15809, IS 15298, IS 3521). The multimodal AI analyzes image pixels to verify: (1) Hard Hats, (2) High-Vis Reflective Vests, (3) Steel-Toe Boots, and (4) Fall-Arrest Harnesses.



Figure 1(a): 100% Compliant Scaffolding Crew
Safety Score: **94/100** (All mandatory PPE verified)



Figure 1(b): Critical Site Hazards Detected
Safety Score: **42/100** (Unshored trench, missing helmets)

Verified Output JSON Schema (PPE Compliance Contract):


```
{ "safety_score": 94, "overall_status": "COMPLIANT", "audit_timestamp": "2026-10-29T18:00:00Z",
"workers_detected_count": 2, "standards_referenced": ["OSHA 1926.100", "IS 2925", "IS 15809", "IS 15298", "IS 3521",
"IS 456"], "ppe_detection": [ {"item": "Industrial Safety Helmets", "required_by": "IS 2925", "detected": true,
"compliant": true, "compliance_rate_pct": 100}, {"item": "High-Visibility Reflective Vests", "required_by": "IS
15809", "detected": true, "compliant": true, "compliance_rate_pct": 100}, {"item": "Steel-Toe Safety Footwear",
"required_by": "IS 15298", "detected": true, "compliant": true, "compliance_rate_pct": 100}, {"item": "Full-Body
Safety Harnesses", "required_by": "IS 3521", "detected": true, "compliant": true, "compliance_rate_pct": 95} ],
"identified_site_hazards": [ {"hazard": "Scaffold coupler retorque required", "severity": "Low", "location": "Grid
B-2", "corrective_remedy": "Retorque coupler bolts to 40 Nm"} ], "executive_summary": "Superb site safety adherence
conforming strictly to IS 456 and OSHA 1926." }
```

4. Tab 2: Concrete Crack Diagnostics & Structural Integrity (Epic 2)

Under **IS 456:2000 Clause 35.3.2**, crack widths in reinforced concrete must not exceed **0.3 mm** for moderate environments, and **0.2 mm** for severe coastal exposure. The diagnostic engine classifies defect types, measures aperture width in mm, identifies engineering root causes, and prescribes remedial repair procedures.



Figure 2(a): 45° Diagonal Structural Shear Crack Width: **6.5 mm** | Severity: **Critical** | Urgency: **Immediate (24h)**



Figure 2(b): Superficial Plaster Hairline Crack Width: **0.3 mm** | Severity: **Minor** | Urgency: **Monitor (30d)**

Table 2: Concrete Defect Classification & Remediation Protocols (IS 456 & ACI 224R)

Defect Pathology	Aperture (mm)	Integrity Risk	Engineering Repair Protocol	Est. Cost (INR)
Structural Shear	3.0 – 10.0 mm	Critical (Imminent)	Install Acro-prop shoring within 24h; Low-viscosity epoxy pressure injection (Sikadur-52) @ 2-4 bar; CFR	■1,400 – ■1,900 / m
Hairline Shrinkage	0.1 – 0.4 mm	Low (Non-structural)	V-groove to 2mm, elastomeric synthetic resin micro-filler (Dr. Fi	■80 – ■120 / m² coats anti-carbonation acry
Foundation Settlement	2.0 – 8.0 mm	High (Progressive)	Geotechnical underpinning, high-pressure polyurethane soil gr	■3,500 – ■6,000 / inch pins with epoxy mo
Thermal Expansion	1.0 – 3.5 mm	Moderate	Rake out failed joint; install closed-cell PE backing rod; apply p	■250 – ■550 / l elastomeric joint sealant.
Spalling & Corrosion	10 – 35 mm	Severe (Cover Loss)	Chisel to sound concrete; sandblast rebar; apply zinc-rich prime	■450 – ■850 / sq.ft micro-concrete re-gro

5. Tab 3: Planning, Real Estate Valuation & Material Engine (Epic 3)

The econometric valuation engine calculates project budgets, CapEx incurred, and market valuations in **INR (Lakhs & Crores)** across Tier 1, Tier 2, and Tier 3 Indian cities based on Central Public Works Department (CPWD) benchmarks.

Mathematical Econometric Formulation:

1. Unit Construction Cost: Rate = BaseRate_struct × TierMultiplier × QualityMultiplier
2. Total Project Budget: Budget_INR = Rate × BuiltUp_sqft
3. Capital Expenditure Incurred: CapEx = Budget_INR × (CompletionPercentage / 100)
4. Projected Ready Market Value: ReadyVal = Budget_INR × MarketMultiplier_tier
5. Current Work-in-Progress (WIP) Valuation: WIPVal = (ReadyVal × LandShare%) + (CapEx × 1.25)

Table 3: Material Consumption Forecasting (IS 456 / CPWD Thumb Rules)

Material	IS Standard	Empirical Norm per sq.ft	Benchmark Case (35,000 sq.ft)	Consumed @ 45% Comp
Cement	IS 12269 (53-G OPC)	0.42 bags / sq.ft	14,700 Bags (735.0 Metric Tons)	6,615 Bags (330.8 MT)
TMT Rebar Steel	IS 1786 (Fe500D)	3.80 kg / sq.ft	133.0 Metric Tons	59.85 Metric Tons
Coarse Sand (M-Sand)	IS 383:2016	1.85 cu.ft / sq.ft	647.5 Brass (64,750 cu.ft)	291.4 Brass (29,140 cu.ft)
Coarse Aggregate (20mm)	IS 383:2016	2.30 cu.ft / sq.ft	805.0 Brass (80,500 cu.ft)	362.3 Brass (36,230 cu.ft)

Conceptual 3D Architectural Rendering Integration (Pollinations AI / Flux)

MCIP integrates an automated architectural prompt synthesizer interfacing with the Pollinations AI / Flux API. It generates 1024x768 photorealistic 3D visual concepts in real time across styles such as *Sustainable Biophilic* (vertical gardens, solar pergolas), *Parametric Glass Curtain Walls*, and *Contemporary Indian Urban* (terracotta jali screens).

Sample Synthesized Prompt: Masterpiece 3D architectural rendering of a Commercial Office Park. Design features lush vertical gardens, hanging green balconies, solar pergolas. Eye-Level Street Hero Perspective, illuminated by Golden Hour Sunset. Rendered in Unreal Engine 5, Octane render, photorealistic 8k.

6. Tab 4: Weather-Aware Advisory & Engineering Assistant (Epic 5)

Severe weather causes site accidents and concrete failures. The Weather Advisory module maps meteorological conditions to operational directives under OSHA 1926.1431 and Indian Standard IS 7861 (Hot & Cold Weather Concreting):

Weather Condition	Severity	Tower Crane Directives	IS 456 Curing & Concreting Protocol	Electrical Safety
High Wind (>35 km/h)	High Alert (Red)	MANDATORY HALT: Suspend all crane lifts & hoists (OSHA 1926.1431). Free-slew cranes and apply anti-rotation brakes.	Halt all heavy concreting operations; cover exposed fresh concrete with heavy tarpaulins to prevent wind-induced surface cracking.	Ensure all temporary site lighting towers and scaffolding are properly earthed.
Heavy Rain / Monsoon	Critical (Blue)	RESTRICTED: Operating surfaces must have anti-skid plates. Secure unbalanced loads and ensure operator cabins are sealed.	HALT FRESH POURS: Cover vertical and horizontal surfaces with polythene sheets. Implement rapid drainage to prevent water ponding.	Check all RCCB/ELCB protection systems for moisture ingress before restarting machinery.
Extreme Heat (>40°C)	Moderate (Amber)	OPERATIONAL: Verify cabin AC; mandatory 10-minute cooling breaks every 2 hours.	HOT WEATHER CONCRETING (IS 7861 Part 1): Monitor concrete temperature; use chilled mixing water or ice to maintain placing temperature below 40°C.	Use insulated gloves when handling metal tools or touching steel reinforcement bars.
Normal Favorable	Normal (Green)	STANDARD: Conduct pre-lift inspections and load charts as per OSHA 1926.1431.	STANDARD: 7-day wet curing (ponding) for OPC concrete; 14 days for PPC/Fly Ash (IS 456 Clause 13.5).	Standard PPE compliance (helmet, boots, vest) required for all personnel on site.

SiteVision AI Engineering Assistant Chatbot

SiteVision AI is an interactive multi-turn conversational assistant grounded in civil engineering codal provisions. It accurately resolves queries regarding **IS 456 Table 11** (Formwork stripping times: 16-24h for vertical columns, 7-14 days for slab props, 14-21 days for beam props) and **IS 456 Table 16** (Nominal clear cover: 50mm footings, 40mm columns, 25mm beams).

■ **Site Engineer:** What is the minimum stripping time for beam props spanning 5.5 meters under IS 456 Table 11?
■ **SiteVision AI:** Under **IS 456:2000 Table 11 (Clause 11.3)**, for props under beams and arches spanning up to 6 meters using Ordinary Portland Cement (OPC), the minimum stripping time is **14 days** (increased to 21 days for spans over 6 meters). If Portland Pozzolana Cement (PPC) or mineral admixtures (fly ash, GGBS) are used, stripping must be prolonged until concrete cube test specimens achieve at least 85% of characteristic compressive strength (fck).

7. Complete Modular Source Code Architecture

The codebase is structured into modular, production-ready components with zero external database dependencies:

7.1 Multimodal Vision Client Engine (modules/gemini_client.py)

```
class GeminiMultimodalClient: BASE_URL = "https://generativelanguage.googleapis.com/v1beta/models" def __init__(self, api_key: Optional[str] = None, model: str = "gemini-1.5-flash"): self.api_key = (api_key or "").strip() self.model = model or "gemini-1.5-flash" def analyze_image_with_json_contract(self, image: Image.Image, system_prompt: str, user_prompt: str, demo_fallback_data: Dict[str, Any]): if self.is_demo_mode: return demo_fallback_data, True, "Demo Mode active. Civil simulation loaded." endpoint = f"{self.BASE_URL}/{self.model}:generateContent?key={self.api_key}" img_b64 = self._pil_to_base64_jpeg(image) payload = { "contents": [{"role": "user", "parts": [ {"text": f"{system_prompt}\n\nTask: {user_prompt}\nRespond ONLY with valid JSON conforming to schema."}, {"inline_data": {"mime_type": "image/jpeg", "data": img_b64}} ]}], "generationConfig": {"temperature": 0.2, "response_mime_type": "application/json"} } response = requests.post(endpoint, headers={"Content-Type": "application/json"}, json=payload, timeout=45) if response.status_code == 200: parsed = self._clean_json_response(response.json()["candidates"][0]["content"]["parts"][0]["text"]) return parsed, False, "Live Gemini 1.5 Multimodal Analysis Complete" return demo_fallback_data, True, "API error -> Fallback simulation loaded."
```

7.2 Econometric Valuation Engine (modules/valuation.py)

```
def calculate_project_financials(structure_type, built_up_sqft, location_tier, completion_pct, quality_spec): base_rate = struct_info["base_rate_per_sqft"] tier_mult = tier_info["base_cost_multiplier"] effective_rate = base_rate * tier_mult * quality_multiplier total_budget = effective_rate * built_up_sqft capex_incurred = total_budget * (completion_pct / 100.0) projected_ready_market_value = total_budget * tier_info["market_val_multiplier"] base_land_value = projected_ready_market_value * (tier_info["land_share_pct"] / 100.0) current_asset_valuation = base_land_value + (capex_incurred * 1.25) # IS 456 Material Forecasting Thumb Rules cement_bags = built_up_sqft * struct_info["cement_bags_per_sqft"] steel_mt = (built_up_sqft * struct_info["steel_kg_per_sqft"]) / 1000.0 sand_brass = (built_up_sqft * struct_info["sand_cft_per_sqft"]) / 100.0 aggregate_brass = (built_up_sqft * struct_info["aggregate_cft_per_sqft"]) / 100.0 return {"total_budget_inr": total_budget, "capex_incurred_inr": capex_incurred, "current_asset_valuation_inr": current_asset_valuation, ...}
```

7.3 Repository File Map

- app.py: Streamlit dashboard, custom industrial styling, responsive tabs.
- config.py: City tiers, CPWD cost indices, IS 456 thumb rules, and weather protocols.
- modules/ppe_audit.py: Multimodal PPE compliance prompt & JSON contract engine.
- modules/crack_diagnostics.py: Defect classification, aperture measurement, and CFRP repair specs.
- modules/pollinations_client.py: Parametric 3D architectural rendering integration.
- modules/weather_advisory.py: Meteorological safety rules and dynamic alerts.
- modules/chatbot.py: Civil engineering assistant with IS 456 knowledgebase.

8. Automated Verification Test Suite & Results

To ensure production resilience, the platform includes a rigorous 7-stage automated verification suite (`test_platform.py`). All test suites executed with 100% pass rate:

Test Suite ID	Component Evaluated	Test Criteria	Execution Status
Stage 1	Module Imports	Verify dependencies (config, ppe, crack, valuation, pollinations, cad)	PASSED (100%)
Stage 2	PPE Audit Engine	Verify safety score computation (0-100), hazard flagging & IS 2925 specs	PASSED (Score: 42, Hazards: 3)
Stage 3	Crack Forensics	Verify aperture width, IS 456 Cl. 40 shear classification, and CFRP specs	PASSED (Width: 6.5mm, Immediate)
Stage 4	Valuation & Materials	Validate INR formatting, CapEx calculations, and TMT steel / cement tonnage	PASSED (Budget: ■15.00 Cr, Steel: 190 MT)
Stage 5	Weather Advisory	Verify OSHA 1926 crane halt triggers and IS 7861 hot weather curing	PASSED (Risk Index: 85 & 92)
Stage 6	Civil Engineering Chatbot	Test IS 456 Table 11 formwork stripping times and Table 16 cover specs	PASSED (Domain responses verified)
Stage 7	Pollinations 3D Renderer	Validate URL generation, Flux prompt encoding, and seed randomization	PASSED (Valid 1024x768 endpoint)

9. Conclusion & Academic Significance

The **Multimodal Construction Intelligence Platform** demonstrates the practical application of Generative AI in infrastructure and real estate decision support. By synthesizing multimodal vision processing, civil engineering codes (IS 456 and OSHA 1926), and econometric modeling, the system delivers an intelligent decision-support platform for construction managers.

The application has been engineered to adhere strictly to production standards, featuring zero external database dependencies, graceful degradation to high-fidelity demonstration simulations, and seamless 1-click deployment readiness for Streamlit Community Cloud.

Candidate's Declaration:

Sulabh Kumar Sharma
Enrollment No: **EN23CS304074**
B.Tech Computer Science & Engineering
Medi-Caps University, Indore

Faculty Guide Approval:

Prof. Shruti Mam
Professor of Generative Artificial Intelligence
Department of Computer Science & Engineering
Medi-Caps University, Indore